

Lab #7 - Partially Filled Parallel Arrays and Selection Sort

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Write a C program that reads student names and exam grades from the user. Since the number of students is not known in advance, your program must use partially filled arrays. Input should continue until the user enters a sentinel value to indicate that there is no more data.

Store the data using:

- two parallel arrays:
 - one array for student names
 - one array for student grades
- one pointer array storing the addresses of the names

After all input is completed, first print the student list in the original input order. Then sort the student records by grade in descending order using the selection sort algorithm, and print the sorted list.

While sorting the student names, do **not** directly reorder the original names array. Instead, **use the pointer array**. During sorting, swap the pointers in this array so that the original names array remains unchanged.

Assume that:

- the sentinel value is "END"
- grades are doubles
- each student name is a single word

Sample Execution

Enter student names and gpas.
Type "END" as the student name to stop.

Enter student name: Ali
Enter gpa: 34.3

Enter student name: Zeynep
Enter gpa: 89.5

Enter student name: Hakan
Enter gpa: 90.1

Enter student name: Selin
Enter gpa: 87.9

Enter student name: END

Sorted student list:

Name	GPA
Hakan	90.10
Zeynep	89.50
Selin	87.90
Ali	34.00

```

#include <stdio.h>
#include <string.h>

#define MAX_STUDENTS 100
#define MAX_NAME_LENGTH 10
#define SENTINEL "END"

int main() {
    /* TODO 1: Declare the necessary variables for this program.

    You must declare the following arrays with these names:
    - names      : a 2D char array for student names
    - namePtrs   : a pointer array for student names
    - gpas.      : an integer array for GPA

    You should also declare:
    - a variable to keep track of the number of valid entries
    - any loop/index variables you need
    - any temporary variables you may need during sorting/swapping */

    printf("Enter student names and gpas.\n");
    printf("Type \"%s\" as the student name to stop.\n\n", SENTINEL);

    /* TODO 2: Read student names and gpas until the sentinel value is entered
    or the arrays become full.

    Pseudocode:
    1. Ask the user to enter a student name.
    2. Read the name into the names array.
    3. If the entered name is the sentinel value, stop input.
    4. Ask the user to enter the gpa.
    5. Read the grade into the gpas array.
    6. Store the corresponding pointer to that student name in the pointer array.
    7. Increase the number of valid entries by 1. */

    /* TODO 3: Sort the entered students by gpas in descending order
    using selection sort. */

    /* TODO 4: Print the sorted student list. */
    printf("Sorted student list:\n");
    printf("%-10s %-5s\n", "Name", "GPA");
    printf("%-10s %-5s\n", "----", "----");

    return 0;
}

```